

Han Wu

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EDUCATION

University of Minnesota

Computer Science, Economics, Bachelor of Arts

Aug. 2024 – Dec. 2026

Minneapolis, MN, USA

University of Michigan

Computer Science, Master of Science

Admitted, Incoming

Ann Arbor, MI, USA

WORK EXPERIENCE

Software Engineer Intern

Anyura Inc.

May 2026 – Present

Sunnyvale, CA

- Developed CRM interfaces using **TypeScript**, **React**, and **Vite**, implementing reusable page components, form workflows, and business interactions.
- Built routing and access-control workflows with **React Router** for customer management, appointments, payment records, and notification settings.
- Developed backend services using **NestJS** and **Fastify**, designing APIs for appointments, customer management, notifications, and permission verification.
- Designed and maintained backend infrastructure using **PostgreSQL on AWS RDS** for relational data and **AWS S3** for secure media and object storage.

Software Engineer Intern

Meituan

May 2025 – Aug. 2025

Beijing, China

- Developed merchant-facing web modules using **TypeScript**, **React**, and **Next.js**, independently delivering a payment information section to improve navigation and content discoverability.
- Contributed to an internal pipeline that processed merchant and transaction data from **AWS RDS PostgreSQL** and transformed structured records into vector embeddings for an internal AI agent.
- Built data cleaning and metadata normalization logic for payment datasets, improving agent-assisted merchant data review workflows by **40%**.
- Debugged API and data inconsistencies across the **Next.js** frontend, backend services, and internal AI workflow to improve system integration reliability.

RESEARCH EXPERIENCE

Research Assistant

Heller-Hurwicz Economics Institute, University of Minnesota

Apr. 2025 – Oct. 2025

Minneapolis, MN, USA

- Used **Python** and **Stata** to clean, merge, and analyze large-scale administrative datasets on healthcare access, provider competition, and Medicaid participation across U.S. states.
- Built automated workflows for regression models and summary statistics to evaluate public policy mechanisms.
- Presented findings through poster sessions and public seminars, strengthening academic communication and data storytelling.

Research Assistant

University of Minnesota Libraries

Mar. 2025 – Jul. 2025

Minneapolis, MN, USA

- Conducted a literature review and organized **50+ papers** on crowdsourcing psychological interventions to synthesize current research trends and inform the study design.
- Designed a research questionnaire and helped establish and refine the study's core research framework.
- Collected and analyzed student feedback on the use of digital technologies and the existing library homepage, translated findings into actionable recommendations, and co-authored a research paper.
- Contributed research findings that informed the subsequent update of the University of Minnesota Libraries homepage.

PUBLICATIONS

Yuzhen Lin, Ivonne Xu, Qi Hu, Ge Zhang, Yajin Li, Yuan Gao, and **Han Wu**. **ProtoHGC: Heterogeneous Graph Contrastive Learning with Prototype-Regularized Classification for Transaction Fraud Detection**. Manuscript under review.

L. Costello, K. McManus, and **Han Wu**. **Investigating Proactive Chat: Exploring Potential Users and Accessibility**. *RUSQ: A Journal of Reference and User Experience*, vol. 61, no. 4, pp. 38–52, 2026. [doi:10.5860/rusq.61.4.8786](https://doi.org/10.5860/rusq.61.4.8786).

Hanlu Zhang, Yuan Gao, Ruijie Zhu, **Han Wu**, Shicheng Zhou, and Jiangtian Pan. **Dynamic Attention-Guided Video Generation from Text with Multi-Scale Synthesis and LoRA Optimization**. *ISCATI 2025, IEEE*, pp. 313–316, March 2025.

PROJECTS

AgentForge – Multi-Agent Software Engineering System | *Python, LangGraph, FastAPI, PostgreSQL, Docker*

- Built an event-driven multi-agent workflow using **LangGraph**, orchestrating PM, Coder, Reviewer, and Tester agents through shared state and conditional graph routing.
- Designed an iterative code-generation pipeline that converted natural-language requirements into structured PRDs and code changes, routing failed reviews and tests back to the Coder Agent for automated revision.
- Integrated **Ruff**, **mypy**, and **Pytest** for automated code validation inside isolated **Docker** containers, with PostgreSQL-backed checkpoints for persistent workflow state.

Healthcare Data Processing & Research API | *Python, FastAPI, PostgreSQL, Redis, Docker*

- Built a backend data processing system for large-scale healthcare administrative datasets, developing automated **Python/Pandas ETL** workflows that reduced manual preprocessing time by **60%**.
- Designed normalized **PostgreSQL** schemas and **FastAPI RESTful APIs** for retrieving provider metrics, policy variables, regression summaries, and state-level healthcare indicators.
- Added **Redis** caching and **Pytest** validation to improve query performance and pipeline reliability, and containerized the backend with **Docker** for reproducible deployment.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, C++

Frameworks & Libraries: React, Next.js, Node.js, NestJS, FastAPI, LangGraph, Vite, Tailwind CSS, Pandas

Data & Research Tools: Stata

Databases: PostgreSQL, Redis

Developer Tools: Git, Docker, Docker Compose, Prisma ORM, Railway

Cloud & Services: AWS (EC2, S3, RDS, SES), Twilio

APIs & Testing: RESTful APIs, tRPC, WebSockets, Pytest, Vitest, Playwright